

gdTAI V4.5 development report

Positive-domain expansion, cytotoxic-feature ablation, dual operating modes | 19 August 2026

Executive decision

Development succeeded; promotion is blocked. Receptor context outperformed the restored cytotoxic panel under the frozen five-source criterion, but no genuinely new untouched test cohort exists. V3 balanced remains the deployed default.

The selected candidate reached macro source-heldout F1 **0.843** and macro recall **0.844**. GDTlung recall remained only **40.5%**, so broader positive training reduced but did not eliminate source dependence.

What V4.5 tested

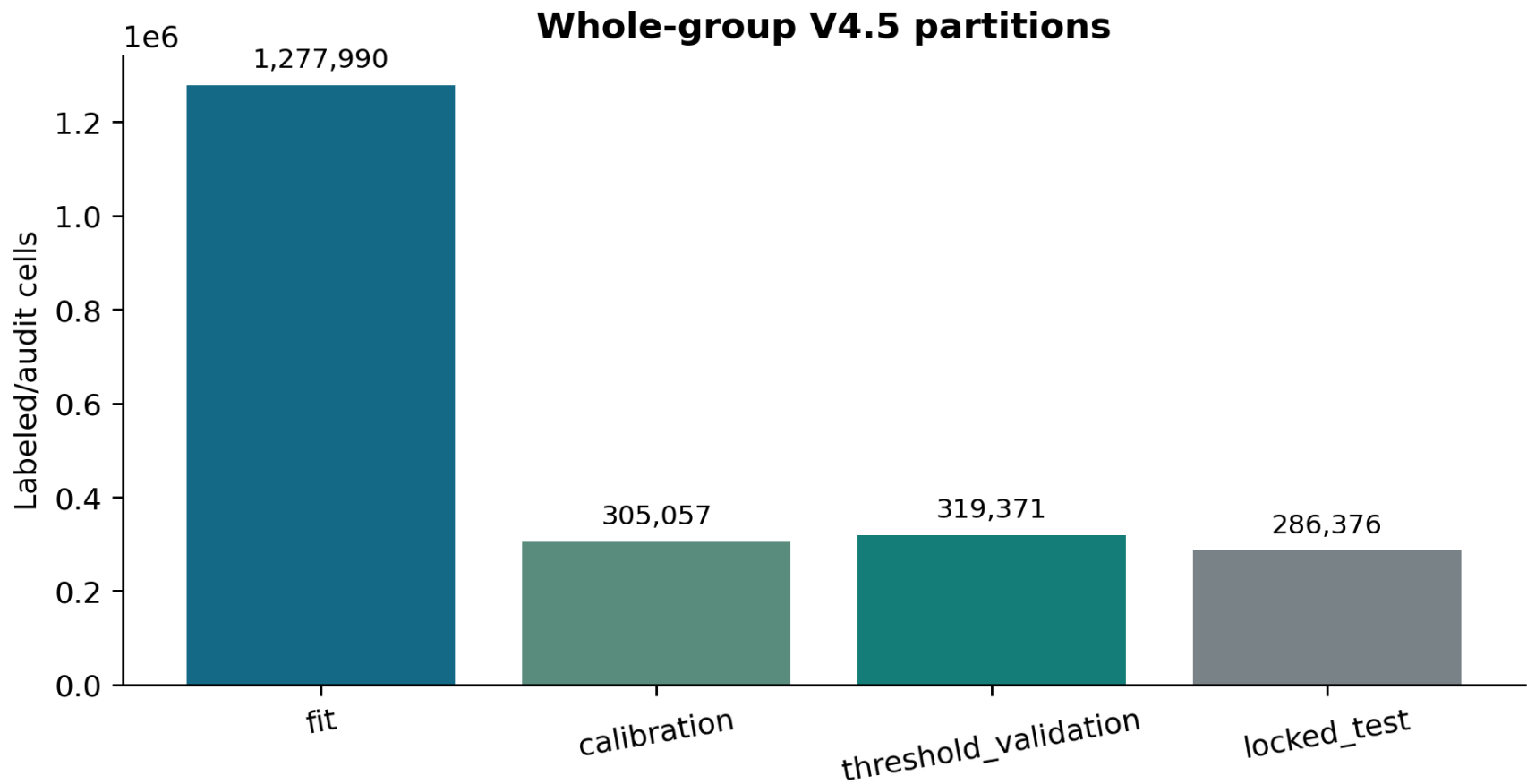
Scientific changes

- Added sorted GDT2020 and GDTlung positives to development with source weights 0.60 and 0.35.
- Kept silver positives out of fitting, calibration, candidate selection, and thresholds.
- Compared 200-feature receptor context against a bounded cytotoxic-context alternative.
- Used curated tier-1 and tier-2 NK negatives; no single NK gene is a hard exclusion.
- Retained the frozen CD4/Treg rules and no hard TRDV-expression cutoff.

Critical validity boundary

GDT2020 and GDTlung came from the already-consumed common lockbox. They are now development data and can never again count as external validation. BALF and extension alpha-beta cohorts were not used by V4.5, but their previous inspection also means the project still needs a new final test.

Ground truth and partitions

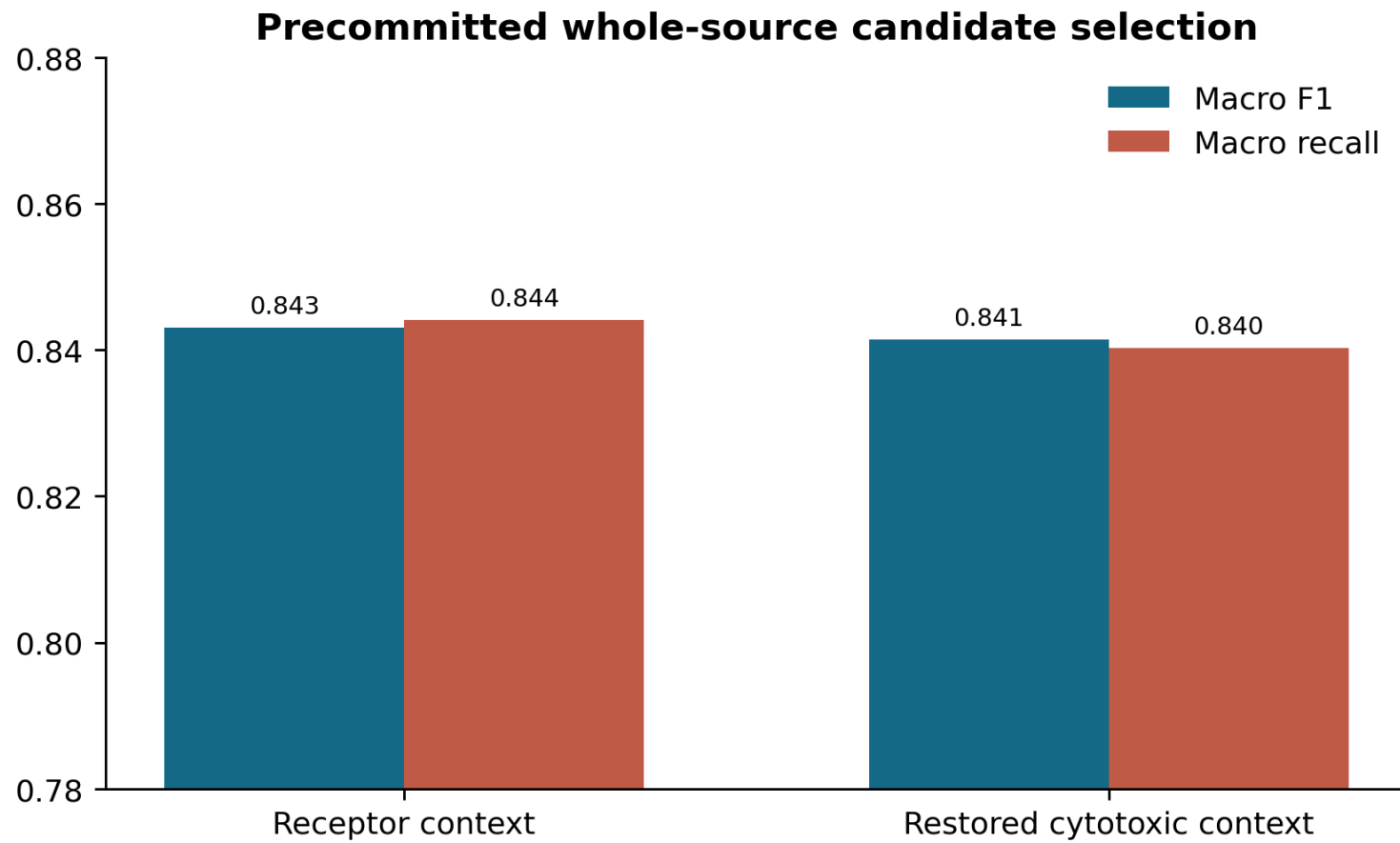


Every biological group stays in exactly one partition. Locked rows do not fit models or calibrators and cannot select features or thresholds.

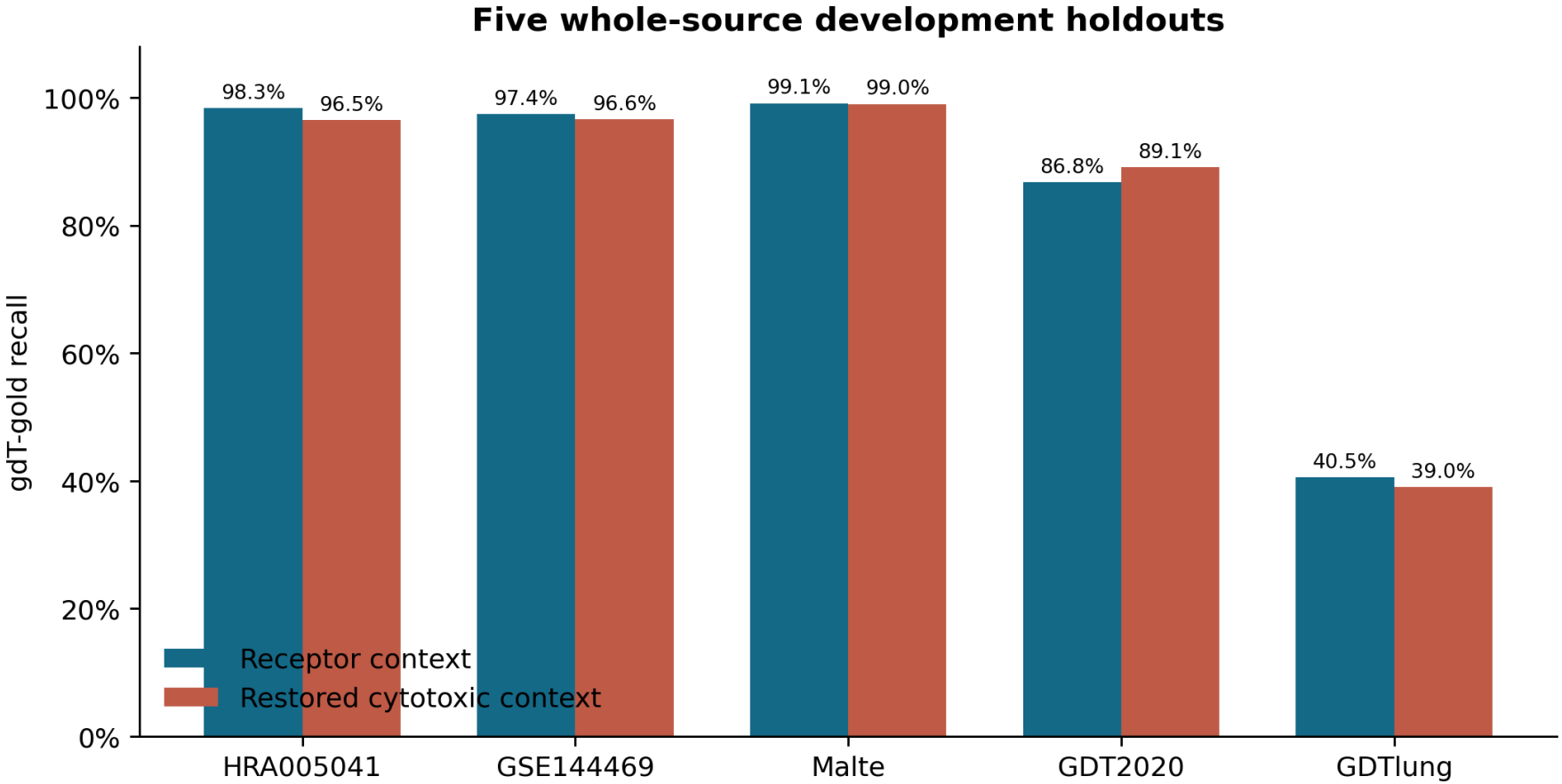
Partition	Positive source	gdT gold
calibration	GDTlung2023july_7p	3,196
calibration	GSE144469	587
fit	GDT_2020AUG_woCOV	16,959
fit	GDTlung2023july_7p	9,296
fit	GSE144469	2,461
fit	HRA005041	2,944

Partition	Positive source	gdT gold
fit	MalteGDT	7,800
locked_test	BALF_BLOOD_COPD	852
threshold_validation	GDT_2020AUG_woCOV	8,945
threshold_validation	GDTlung2023july_7p	2,683
threshold_validation	GSE144469	752
threshold_validation	HRA005041	1,259

Preflight passed. Fit/calibration/threshold-validation contain 47,460/3,783/13,639 gdT-gold cells. No silver cell entered development.

Feature ablation

Candidate selection used macro F1 across five independently refitted whole-source holdouts.



Restored shared cytotoxic genes did not improve aggregate transfer and reduced the weakest-source recall.

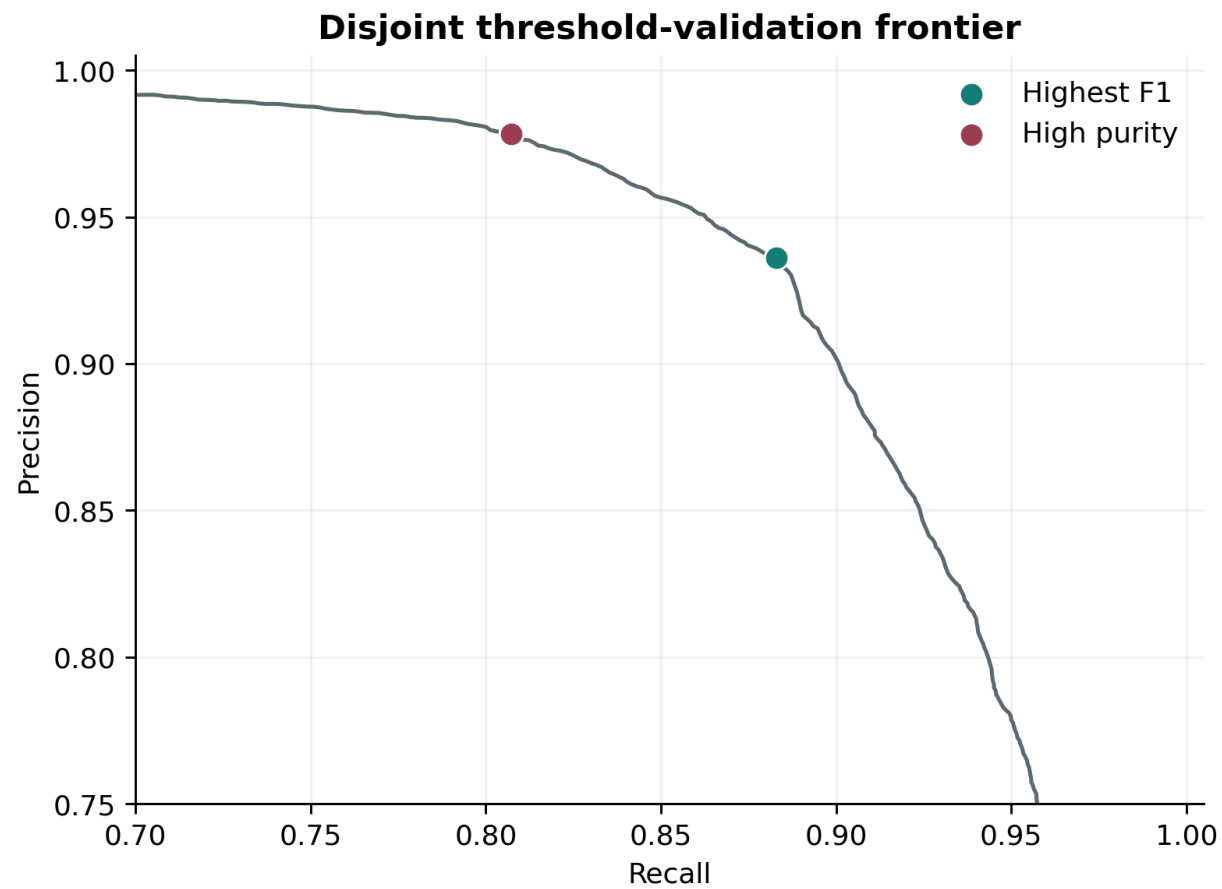
Candidate	Macro F1	Macro recall	Minimum recall	Macro alpha-beta FPR	Selected
Receptor context	0.8431	84.41%	40.51%	0.32%	1
Restored cytotoxic context	0.8414	84.02%	39.02%	0.32%	0

Whole-source failure modes

Candidate	Held-out source	gdT recall	precision	f1	alpha-beta FPR
Receptor context	HRA005041	98.31%	58.30%	0.7319	0.58%
Receptor context	GSE144469	97.37%	99.09%	0.9822	0.07%
Receptor context	MalteGDT	99.09%	100.00%	0.9954	-
Receptor context	GDT_2020AUG_woCOV	86.76%	100.00%	0.9291	-
Receptor context	GDTlung2023july_7p	40.51%	100.00%	0.5767	-
Restored cytotoxic context	HRA005041	96.48%	59.01%	0.7323	0.56%
Restored cytotoxic context	GSE144469	96.55%	98.76%	0.9764	0.09%
Restored cytotoxic context	MalteGDT	98.97%	100.00%	0.9948	-
Restored cytotoxic context	GDT_2020AUG_woCOV	89.07%	100.00%	0.9422	-
Restored cytotoxic context	GDTlung2023july_7p	39.02%	100.00%	0.5614	-

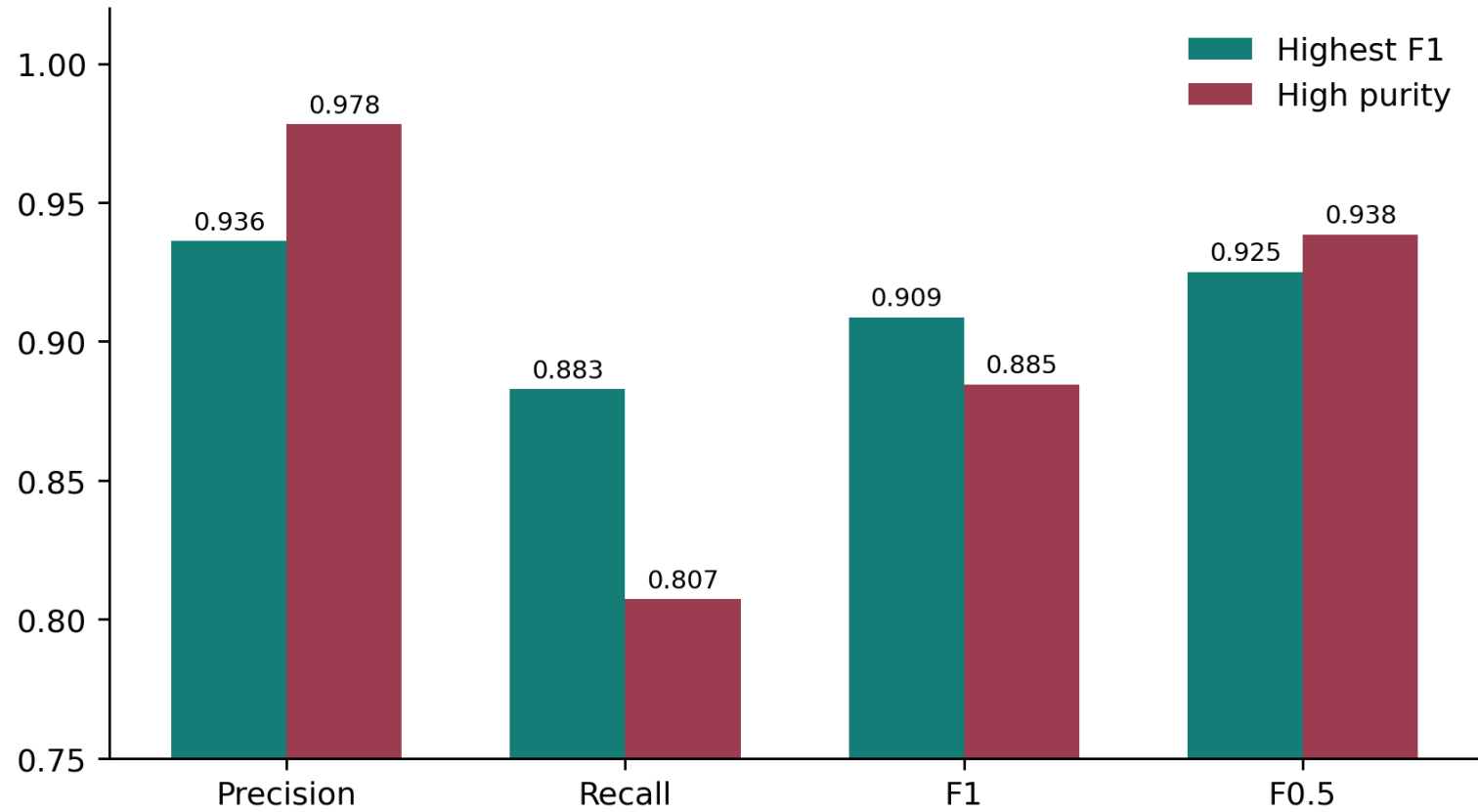
HRA precision is lower because its held-out set contains 506,985 alpha-beta gold cells, whereas the three sorted-only held-out sources contain no negatives and therefore cannot estimate precision or FPR. The sorted-source F1 values should be read primarily as recall summaries.

Frozen operating modes



Both operating points come only from the disjoint development threshold-validation partition.

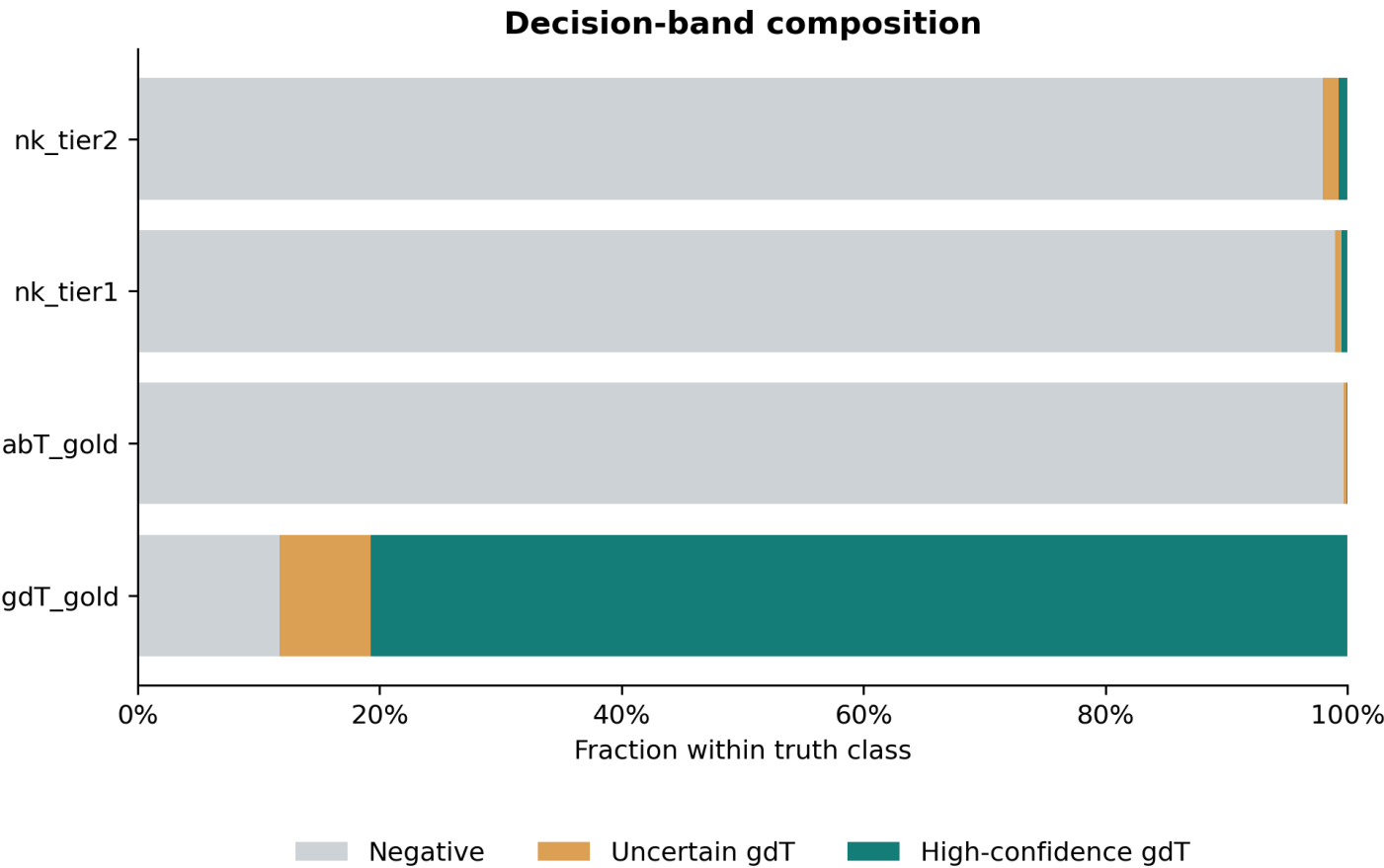
Frozen development operating modes



Highest F1 prioritizes recovery; high purity maximizes F0.5 without a hard FPR constraint.

Mode	threshold	precision	recall	f1	F0.5	specificity
highest_f1	0.8595	93.62%	88.28%	0.9087	0.9250	99.67%
high_purity	0.9657	97.82%	80.72%	0.8846	0.9385	99.90%

Uncertainty band



Scores between 0.859506 and 0.965705 are labeled uncertain, not automatically correct.

Band	Truth	Cells	Within truth
high_confidence_gdt	abT_gold	231	0.09%
high_confidence_gdt	gdT_gold	11,010	80.72%
high_confidence_gdt	nk_tier1	14	0.49%
high_confidence_gdt	nk_tier2	100	0.74%
negative	abT_gold	243,590	99.68%
negative	gdT_gold	1,598	11.72%

Band	Truth	Cells	Within truth
negative	nk_tier1	2,830	98.95%
negative	nk_tier2	13,262	97.98%
uncertain_gdt	abT_gold	560	0.23%
uncertain_gdt	gdT_gold	1,031	7.56%
uncertain_gdt	nk_tier1	16	0.56%
uncertain_gdt	nk_tier2	174	1.29%

The uncertainty interval contains 1,031 gdT-gold and 560 alpha-beta-gold validation cells. It supports conservative review workflows but does not repair low-scoring GDTlung positives below the highest-F1 threshold.

Interpretation

- The V4.3 threshold-transfer defect is addressed because each held-out model and the final deployment scorer are calibrated directly, and final thresholds are derived from that exact scorer.
 - Positive-domain expansion markedly improves held-out HRA, GSE144469, Malte, and GDT2020 recall.
 - The cytotoxic panel is not promoted: NK labels were sufficient to keep its FPR similar, but the added genes provided no reproducible macro-F1 gain.
 - GDTlung remains the dominant domain failure, consistent with its known suboptimal library quality and weak receptor-expression transfer.
 - V4.5 development-validation F1 is not directly comparable to V4.4's easier threshold partition because V4.5 adds difficult consumed sorted cohorts.
 - Do not score the consumed common lockbox again for model selection. Promotion requires a new untouched positive plus alpha-beta/NK test.
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Reproducibility and status

Status: development frozen, not promotable. No H5AD was modified, no whole-atlas inference was run, and no V4 artifact was pushed to GitHub.

Training script: `workflows/gdtai/train_gdtai_v4_5_development.py`

Configuration: `configs/models/gdtai/v4_5_positive_diversity_ablation.json`

Model contract: `Integrated_dataset/models/gdT_prediction/gdtai_v4_5_development/model_contract.json`

Contract SHA-256: `974d0687de4a93688b4e0d665b276458c90f6f5a772615dbeea9cbb2f66072b2`

Runtime: 149.7 seconds after exact vectorization of the unchanged Stage-1 threshold grid.
